

## **SIMATS ENGINEERING**

### **CO4 - ASSESSMENT TOOL 2**

#### **Industry Problem Solving Task**

**COURSE NAME:** Deep Learning

**COURSE CODE:** MLA0408

#### **COURSE OUTCOME COVERED**

**CO 1:** Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. (BL-6)

**Assessment Tool Weightage: 40%**

#### **QUESTIONS:**

**Total Marks:25**

##### **Q1. Transfer Learning**

A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

##### **Q2. DenseNet and PixelNet**

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

##### **Q3. Bidirectional RNN and Encoder–Decoder**

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

##### **Q4. BPTT and LSTM**

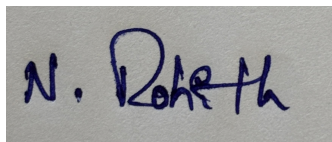
A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

##### **Q5. Integrated Deep Learning Solution**

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

**Code of Conduct:**

I NALIPI REDDY ROHITH/192425115(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:****RUBRICS FOR CALCULATION:**

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |

# Industry Problem Solving Task

## Q1. Transfer Learning – Skin Cancer Detection

A healthcare startup has only a limited number of labeled dermoscopic images for detecting skin cancer. Training a deep CNN completely from scratch would require a large dataset and may lead to overfitting. Therefore, **transfer learning** can be used with a pre-trained CNN such as **ResNet50** or **EfficientNet**.

### Proposed Architecture

**Dermoscopic Image** → **Preprocessing** → **Data Augmentation** → **Pre-trained CNN** → **Global Average Pooling** → **Dense Layer** → **Softmax** → **Benign/Malignant**

First, all dermoscopic images are resized to a fixed size such as **224 × 224 pixels** and normalized. Data augmentation techniques such as horizontal and vertical flipping, rotation, zooming, cropping and brightness variation can increase the effective training data.

A pre-trained **ResNet50** can be used as the feature extractor. The initial layers learn general features such as edges, curves and textures, which are useful for many types of images. Therefore, these layers can be **frozen** during initial training.

The final convolutional layers learn more complex and domain-specific features. These layers can be **fine-tuned** using the skin-cancer dataset with a small learning rate.

The original ImageNet classification layer is removed and replaced with:

- Global Average Pooling
- Fully Connected Layer
- Dropout
- Softmax output layer

The final output can classify images into **benign and malignant** categories.

## Why Transfer Learning?

Transfer learning reduces training time and the amount of labeled data required. The pre-trained CNN already contains useful visual features, so the model only needs to learn the specific characteristics of skin lesions.

### Conclusion:

Transfer learning with selective fine-tuning provides better generalization and reduces overfitting when the medical dataset is small.

## Q2. DenseNet and PixelNet – Road Scene Segmentation

A smart-city road analysis system must identify **roads, pedestrians, vehicles, buildings and traffic signs at the pixel level**. This requires semantic segmentation rather than simple image classification.

A suitable solution is to combine a **DenseNet feature extractor** with a **PixelNet-style pixel-wise classification network**.

### Proposed Architecture

Input Road Image

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DenseNet Backbone

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Dense Connectivity and Feature Reuse

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Multi-level Feature Maps

↓

Upsampling

↓

## **Pixel-wise Classification**

↓

## **Semantic Segmentation Map**

DenseNet uses dense connections where each layer receives feature maps from previous layers. This allows information from early layers to reach deeper layers directly.

For example:

**Layer 1 → Layer 2 → Layer 3 → Layer 4**

In dense connectivity, it becomes:

**Layer 1 → Layer 2, Layer 3, Layer 4**

**Layer 2 → Layer 3, Layer 4**

This improves feature reuse and gradient flow.

The output feature maps are passed to a pixel-wise prediction layer. A **1 × 1 convolution** can generate class scores for every pixel, followed by Softmax to determine the class.

## **Benefits**

### **Feature Reuse:**

Dense connections reuse previously learned features instead of learning the same information repeatedly.

### **Better Segmentation:**

Pixel-level classification helps accurately identify boundaries of cars, pedestrians and traffic signs.

### **Efficient Training:**

Dense connectivity improves gradient propagation and can reduce redundant feature

learning.

### **Improved Accuracy:**

Combining low-level spatial information and high-level semantic information helps produce detailed segmentation maps.

### **Conclusion:**

DenseNet provides strong and reusable image features, while PixelNet-style prediction provides pixel-level classification, making the combined model suitable for autonomous road-scene analysis.

## **Q3. Bidirectional RNN and Encoder–Decoder**

A multinational customer-service company requires a chatbot that can translate queries between multiple languages while preserving their meaning and context.

An **Encoder–Decoder sequence-to-sequence architecture with a Bidirectional RNN** can be used.

### **Proposed Architecture**

**Input Sentence**

↓

**Word Embedding**

↓

**Bidirectional RNN Encoder**

↓

**Context Representation**

↓

**LSTM/RNN Decoder**

↓

## **Target Language Words**

↓

## **Translated Sentence**

The input sentence is first converted into numerical word embeddings. These embeddings are processed by the Bidirectional RNN.

The Bidirectional RNN has two processing directions:

- **Forward RNN:** processes the sentence from left to right.
- **Backward RNN:** processes the sentence from right to left.

Their outputs are combined to create a richer representation.

## **Example**

Suppose the input is:

**"The customer wants to cancel the order."**

The meaning of "cancel" can be better understood by considering both previous and following words.

A unidirectional RNN mainly uses information from previous words, while a Bidirectional RNN can use information from both sides.

## **Encoder**

The encoder converts the complete input sentence into a context representation containing important information about the sentence.

## **Decoder**

The decoder uses the context representation to generate the translated sentence one word at a time.

For example:

**English:**

"Where is my order?"

**French:**

"Ou est ma commande?"

**Advantages**

- Captures both past and future context.
- Improves understanding of sentence structure.
- Helps resolve ambiguous words.
- Provides better contextual representations.
- Suitable for multilingual translation.

**Conclusion:**

Bidirectional processing gives the encoder information from both directions, resulting in better contextual understanding and improved translation accuracy.

#### **Q4. BPTT and LSTM – Stock Price Prediction**

A financial analytics company wants to forecast future stock prices using historical market information. Since stock prices are sequential data, an **LSTM-based time-series model** is suitable.

**Proposed Architecture**

**Historical Stock Data**

↓

**Data Cleaning and Normalization**

↓

**Time-Series Sequence Creation**

↓

**LSTM Layer**

↓

**Dense Layer**

↓

**Predicted Stock Price**

Historical features can include:

- Opening price
- Closing price
- Highest price
- Lowest price
- Trading volume

For example, the previous **10 days** can be used to predict the price on the 11th day.

## **LSTM Working**

LSTM contains a cell state and three main gates:

### **Forget Gate:**

Determines which old information should be removed.

### **Input Gate:**

Determines which new information should be stored.

### **Output Gate:**

Determines which information should be passed to the next layer.

These gates allow the network to maintain useful information over long sequences.

## **BPTT**

During training, the LSTM is conceptually **unrolled across multiple time steps**.

For example:

Day 1 → LSTM



Day 2 → LSTM



Day 3 → LSTM



Day 4 → LSTM



Prediction

The prediction error is calculated using a loss function such as Mean Squared Error.

The error is propagated backward through the time steps. Gradients are calculated and the weights are updated using an optimizer such as Adam.

This process is called **Backpropagation Through Time (BPTT)**.

### **Why LSTM instead of Standard RNN?**

A standard RNN can suffer from vanishing gradients when sequences become long. As a result, it may forget information from earlier time steps.

LSTM uses gates and a cell state to preserve important information for longer periods.

Therefore, LSTM is better suited for financial time-series data where earlier market patterns may influence future prices.

### **Conclusion:**

LSTM with BPTT can learn both short-term and long-term temporal patterns, making it more suitable than a standard RNN for stock-price forecasting.

## Q5. Integrated Deep Learning Solution – Video Caption Generation

A media streaming company wants to automatically generate captions for uploaded videos. The proposed system combines **Transfer Learning, CNN-based visual feature extraction and an LSTM Encoder–Decoder**.

### Overall Architecture

Video



Frame Extraction



Pre-trained CNN



Visual Feature Extraction



LSTM Encoder



Context Vector



LSTM Decoder



Word Prediction



Generated Caption

### 1. Video Frame Extraction

A video contains many frames. Instead of processing the complete video at once, selected frames are extracted at regular intervals.

For example:

Frame 1 → Frame 2 → Frame 3 → Frame 4 → Frame 5

## 2. Transfer Learning

A pre-trained CNN such as **VGG16**, **ResNet50** or **MobileNetV2** is used to extract visual features.

The final classification layer of the CNN is removed because the objective is not ImageNet classification.

The CNN extracts information such as:

- Objects
- People
- Vehicles
- Background
- Shapes
- Visual patterns

Initially, the CNN layers can be frozen to preserve their learned features.

## 3. LSTM Encoder

The CNN produces a feature vector for every video frame.

These feature vectors form a sequence:

F1 → F2 → F3 → F4 → F5

The LSTM encoder processes this sequence and learns temporal relationships.

For example, it can understand:

Person → picks up ball → throws ball

rather than treating each frame independently.

## 4. LSTM Decoder

The decoder receives the encoded information and generates words sequentially.

For example:

<START>

↓

A

↓

boy

↓

is

↓

playing

↓

football

↓

<END>

The decoder predicts the next word based on the previous generated word and the video context.

## 5. Role of Each Component

| Component         | Main Function                            |
|-------------------|------------------------------------------|
| CNN               | Extracts visual features                 |
| Transfer Learning | Uses previously learned visual knowledge |

|                |                                        |
|----------------|----------------------------------------|
| LSTM Encoder   | Learns temporal information            |
| Context Vector | Represents important video information |
| LSTM Decoder   | Generates caption words                |
| Softmax        | Selects the most probable next word    |

### Example

For a video showing a person playing football, the system may generate:

**"A boy is playing football."**

For a video showing a person walking:

**"A man is walking on the road."**

### Advantages

- Reduces the need to train a CNN from scratch.
- Extracts meaningful visual features.
- LSTM captures temporal relationships between frames.
- Encoder–Decoder structure generates captions sequentially.
- Can be extended to different languages.
- Helps improve accessibility for visually impaired users.

### Conclusion:

The integrated CNN–LSTM architecture combines strong visual representation with temporal sequence learning, enabling the system to generate meaningful captions from video content.